MIL-S-6180B
24 September 1964
SUPERSEDING
MIL-S-6180A
22 August 1956

MILITARY SPECIFICATION

SWAGING TOOLS, FLEXIBLE CABLE TERMINAL, HAND-OPERATED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers hand-operated, flexible control cable terminal swaging tools.

1.2 **Classification.** Swaging tools shall be of the following types, as specified (see 6.2):

Type I — Portable, for 1/16 to 1/8-inch ball-type terminals and for 1/16 to 5/16-inch sleeve-type terminals.

Type II — Bench, for 5/32 to 1/4-inch ball-type terminals and for 7/32 to 3/8-inch sleeve-type terminals.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-5688 — Cable Assemblies; Aircraft, Proof Testing and Prestretching of.

MIL-T-45542 — Tool Kits, Shop Sets, and Tool Sets, (Both Common and Special), Preparation For Delivery Of: General Specification For.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151—Metals; Test Methods.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

MS-20658 — Terminal, Wire Rope, Swaging, Fork End.

MS-20663 — Ball End, Wire Rope, Swaging, Double Shank.

MS-20664 — Ball End, Wire Rope, Swaging, Single Shank.

FSC 5120

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MS-20667 — Terminal, Wire Rope, Swaging, Fork End.

MS-20668 — Terminal, Wire Rope, Swaging, Eye End.

MS-21259 — Terminal, Wire Rope, Stud.

MS-21260 — Terminal, Wire Rope, Stud.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

≠ 3.1 **Preproduction sample.** When specified (see 6.2), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 **Materials.** Materials shall be of such quality as to assure conformance to the requirements specified herein.

* 3.3 **Design and construction.** The swaging tools shall be constructed from metal, of the quality normally used in the manufacture of this item, and shall be designed to swage cable terminals to steel, aircraft control cables. The swaged cable terminals shall conform to MS-20658, MS-20663, MS-20664, MS-20667, MS-20668, MS-21259 and MS-21260. The swaged terminals shall be capable of withstanding the tests specified in MIL-C-5688. All movable parts of the swaging tools that are subject to wear shall be hardened and tempered to show a hardness range of Rockwell C56 to C62. Determination of hardness shall be as specified in 4.4.2. The swaging tools shall be provided with built-in cable and terminal mechanisms for guiding and supporting the terminals during the swaging operation.

≠ 3.3.1 *Type I swaging tool.* The type I swaging tool shall be similar to figure 1 and shall be capable of swaging ball-type terminals to cables having 1/16, 3/32, and 1/8-inch diameters; and swaging sleeve-type terminals to cables having 1/16, 3/32, 1/8, 5/32 and 3/16-inch diameters.

= 3.3.2 *Type II swaging tool.* The type II swaging tool shall be similar to figure 2 and shall be capable of swaging ball-type terminals to cables having 5/32, 3/16, 7/32, and 1/4-inch diameters; and swaging sleeve-type terminals to cables having 7/32, 1/4, 9/32, 5/16, and 3/8-inch diameters.

* 3.3.3 *Operational design.* The swaging tool shall be designed so that not more than one operator will be required to operate the tool.

* 3.3.4 *Swaging time and force.* Swaging tools shall be capable of swaging terminals to the dimensions specified on applicable standards (see 3.3) and in accordance with the time and force shown on table I, when tested as specified in 4.4.1.

3.3.5 *Lubrication.* Provisions for lubrication of all applicable movable parts shall be provided on the swaging tools.

3.4 **Gages.** Gages shall be provided with each swaging tool and shall be substantially as shown on figures 1 or 2, as applicable. The gages shall be made of hardened, tempered steel. Measuring surfaces of the gage openings shall be ground finished after hardening and shall be flat and straight over their entire area. The gages shall include both go and no-go openings conforming to the sizes of cable terminals specified in 3.3. Openings shall be capable of measuring the finished swaged diameter sizes of the ball- and sleeve-type terminals specified herein. To identify the individual gage size for the terminal diameters specified herein, the gage openings

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TABLE I. *Swaging time and force*

Type tool	Cable diameter	Force exerted on handle (maximum)		Time to swage (approximate)		
		Ball-type terminal	Sleeve-type terminal	Ball-type terminals		Sleeve-type terminals
				Single shank	Double shank	
	Inch	*Pounds*	*Pounds*	*Seconds*	*Seconds*	*Seconds*
I	1/16	18	16	46	88	30
I	3/32	54	26	46	88	30
I	1/8	96	30	50	96	32
I	5/32	....	56	.....	.....	34
I	3/16	....	74	.....	.....	48
II	5/32	40	.....	35	66	.....
II	3/16	50	.....	45	74	.....
II	7/32	70	35	54	102	60
II	1/4	100	50	64	120	60
II	9/32	....	80	.....	.....	90
II	5/16	....	120	.....	.....	120
II	3/8	....	135	.....	.....	150

shall be marked for size in a legible and permanent manner on one side of the gage plate.

3.5 Carrying case. A metal carrying case of not less than 0.0239-inch (24 gage) thickness shall be provided for each swaging tool and shall be similar to the case shown on figures 1 or 2, as applicable. The carrying case shall be provided with compartments to accommodate loose parts.

3.5.1 *Dimensions.* The overall dimensions of carrying case shall not exceed the following:

Type I Swaging Tool	*Type II Swaging Tool*
Length 31 inches	Length 19 inches
Width 7 inches	Width 10 inches
Height 6⅝ inches	Height 14 inches

3.6 Weight. The weight of the assembled type I swaging tool shall not exceed 28 pounds without the case nor 66 pounds with the case and accessories. The weight of the assembled type II swaging tool shall not exceed 100 pounds without the case nor 150 pounds with the case and accessories.

3.7 Finish. The swaging tools shall be finished in accordance with manufacturer's commercial practice.

3.8 Operating instructions. Operating instructions shall be provided and securely attached to the inside of each carrying case.

3.9 Marking for identification.

* 3.9.1 *Swaging tools.* The swaging tools shall be marked in accordance with MIL-STD-130.

3.9.2 *Carrying case.* Each carrying case shall be permanently and legibly marked with the following information:

Swaging Tool, Flexible Cable Terminal, Hand-Operated Type.

Capacity .....

Sleeve-Type terminals

Ball-Type terminals

Specification No. _____

Contract or Order No. _____

Stock No. _____

Mfr's Part No. _____

Mfr's Name or Trade Mark _____

Letters "U.S." _____

* **3.10 Workmanship.** The finished item shall conform to the quality and grade of product established by this specification. All fins, burrs, or sharp edges shall be removed.

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4. QUALITY ASSURANCE PROVISIONS

= 4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

= 4.2 **Preproduction sample inspection.** When a preproduction sample is required, it shall be examined for defects in table II, dimensions and weights specified, and tested as specified in 4.4.1.

= 4.3 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3.1 *Inspection of materials and components.* In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

= 4.3.2 *Intermediate testing.* Testing for hardness shall be performed on each sample unit prior to application of finish, in accordance with 4.4.2. A minimum of two (2) determinations shall be made on each sample unit and no individual reading may exceed the specified requirements. The sample unit shall be one (1) of each movable part (subject to wear) from each type of swaging tool. A lot shall consist of all movable parts of one type submitted for inspection at one time. The inspection level shall be S-2 with an acceptable quality level of 1.5 defects per hundred units.

= 4.3.3 *Inspection of the end item.* The lot shall consist of all swaging tools of one type submitted for acceptance at one time. The sample unit shall be (1) complete swaging tool.

= 4.3.3.1 *Visual examination.* Visual examination of the end item shall be in accordance with the classification of defects in table II. The inspection level shall be level II, with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

4.3.3.2 *Dimensional and weight examination.* Examination shall be made for compliance with dimensions and weights specified herein. Any deviation therefrom shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.3.3 *Testing of end item.* Testing shall be performed on each sample unit in accordance with 4.4.1. The inspection level shall be level S-3 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

TABLE II. *Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship.	Missing component or wrong type	X	
	Fins, burrs or sharp edges	X	
	Type not as specified	X	
	Provision for lubrication missing	X	
Finish	Not finished as specified		X
Marking	Missing, not legible, incomplete		X
Instructions	Missing, not legible, incomplete	X	

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≠ 4.3.4 *Examination of preparation for delivery.* Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with MIL-T-45542.

4.4 Tests.

* 4.4.1 *Performance test.* The swaging tool shall be tested by swaging one of each size and type terminal to cable sizes listed in table I. Terminals shall be proof tested in accordance with MIL-C-5688. Each terminal shall be examined for compliance with requirements specified in 3.3.4.

≠ 4.4.2 *Hardness test.* Tests shall be performed in accordance with Method 243.1 of Fed. Test Method Std. No. 151. The values thus obtained shall comply with the requirements of 3.3.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Each complete type I or II swaging tool, with corresponding carrying case, shall be preserved, packaged and packed for each level of protection in accordance with the applicable requirements conforming to type I of MIL-T-45542.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

Custodians:
Army—GL
Navy—WP
Air Force—69

Review activities:
Army—MO
Navy—WP
Air Force—69

User activities:
Army—SM
Navy—SM

6. NOTES

6.1 **Intended use.** The swaging tools covered by this specification are intended for use in attaching cable terminals to aircraft control cables.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of swaging tool required (see 1.2).
(c) When preproduction sample is required (see 3.1).
(d) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 The outside margins of this document have been marked to indicate where changes from previous issue have been made. An asterisk (*) indicates a substantive change in paragraph text. A double asterisk (**) indicates deletion of a paragraph bearing the same number. The sign (≠) indicates the addition of a completely new paragraph. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Army—GL
Project No. 5120-0422

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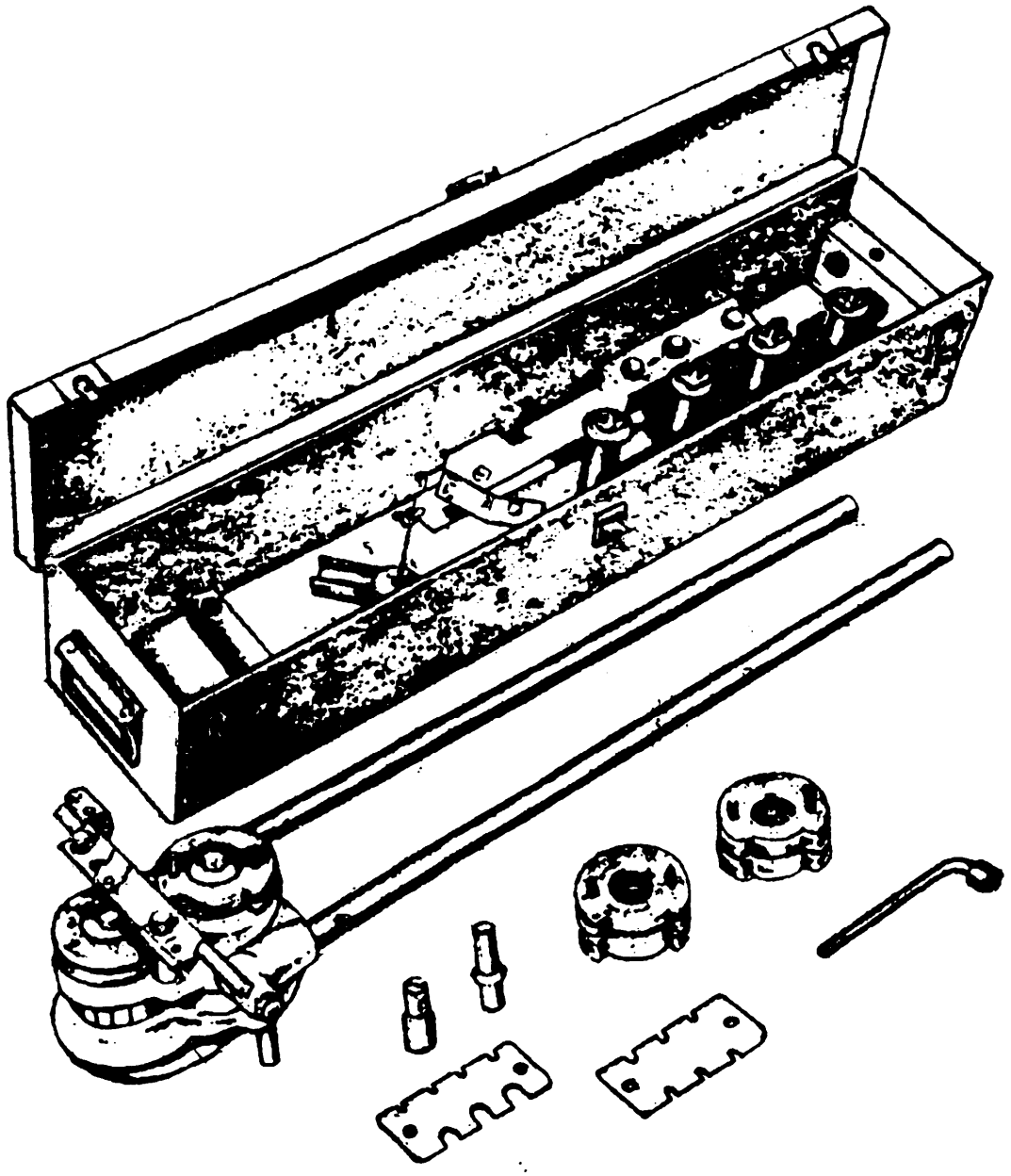


FIGURE 1. *Type I swaging tool.*

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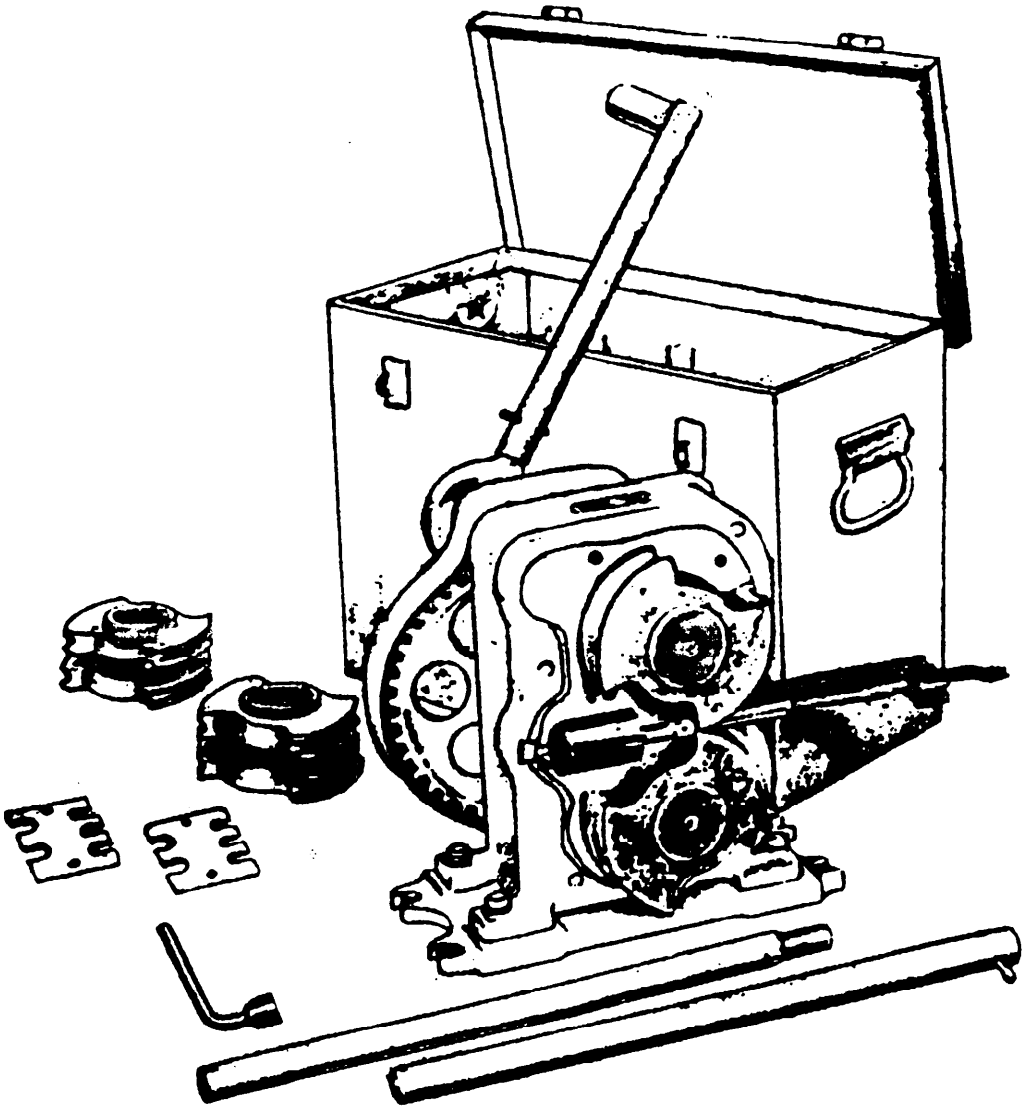


FIGURE 2. *Type II swaging tool.*

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